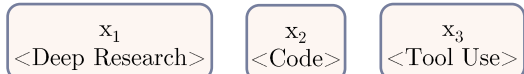
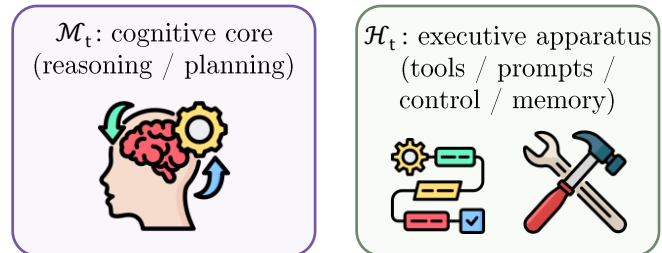


Task Set

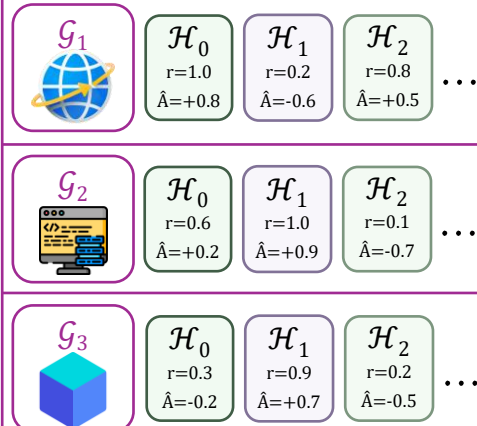


Agent ($\mathcal{M}_t, \mathcal{H}_t$)



Replay Buffer $\mathcal{B}$

Cross-Harness Grouping



$$\mathcal{G}_x = \{\tau \in \mathcal{B} | task(\tau) = x\}, \quad \hat{A}_i = \frac{r_i - \mu_G}{\sigma_G + \epsilon}$$

AEGIS Harness Evolution

Non-parametric optimization of $\mathcal{H}_\theta$



Trajectories



Evolve Agent

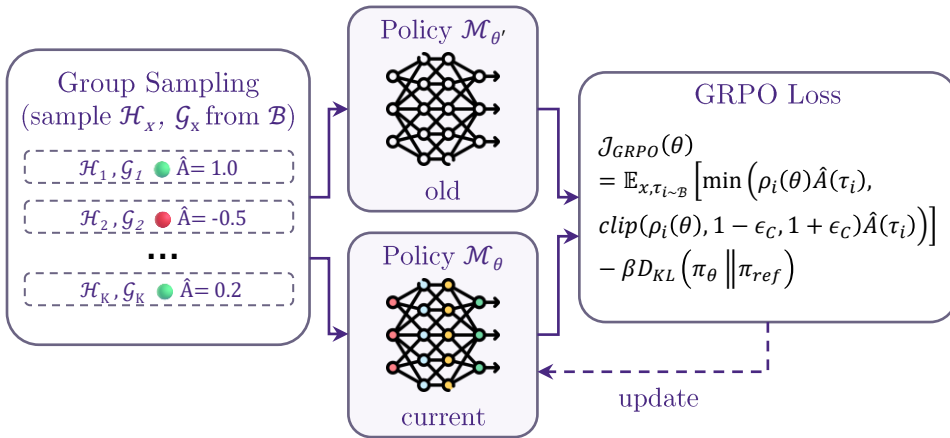


$\mathcal{H}_{t+1}$

evolved harness
+ audit trail

Cross-Harness GRPO

Parametric optimization of $\mathcal{M}_\theta$



$\mathcal{M}_{t+1}$

updated model
policy